

TAO (TONY) JIN

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EDUCATION

Cornell University

Master of Engineering in Systems Engineering

Ithaca, NY
Sep. 2025 – May 2026

University of British Columbia

Bachelor of Science: Major in Mathematics

Vancouver, BC
Sep. 2021 – June 2025

- Relevant Coursework: Statistical Learning, Time Series Analysis, Statistical Inference & Hypothesis Testing, Stochastic Calculus, Linear Programming, Algorithms, Relational Databases.

PROFESSIONAL EXPERIENCE

Aifuku

Data Science Intern

Toronto, ON
Jan. 2025 – May 2025

- Wrote and optimized PostgreSQL queries (joins, CTEs, window functions) on a Postgres 15 source-of-truth database of 9 tables, validating order, receipt, inventory, and ledger data feeding Metabase business intelligence (BI) dashboards and KPI reporting; ran exploratory data analysis (EDA) and root cause analysis across 5 linked entities, presented findings and recommendations cross-functionally to non-technical stakeholders, lifting data quality and reporting accuracy.
- Built and documented a Mage ETL pipeline pulling statement PDFs from Google Drive, parsed transaction fields with a Gemini Large Language Model (LLM) API and prompt engineering, and output structured JSON for ledger reconciliation, replacing manual data entry with reproducible automation.

Grow Investment Group

Institutional Sales Department Intern

Shanghai, China
May 2024 – Aug. 2024

- Engineered Python (pandas) automation scripts cross-referencing multi-source databases for institutional client due diligence and due diligence questionnaire (DDQ) completion, cutting manual data processing time 15% through systematic data validation.
- Authored convertible bond downturn analysis, quantitative market research on macroeconomic, rate, and sentiment data, and product training materials, translating findings into data storytelling for non-technical audiences; drove 9% fewer redemptions, 4% more inquiries, 12% higher conversion.

Sinolink Securities Co., Ltd.

Investment Banking Intern

Shanghai, China
June 2023 – Aug. 2023

- Calculated portfolio risk KPIs (Sharpe, Sortino, Beta, volatility) and correlation analysis for stakeholder risk management reports; ran pre-IPO due diligence (~RMB 460M) with rigorous attention to detail and redesigned reporting workflows, improving team efficiency 8%.

PROJECT EXPERIENCE

Marketplace Retention & Fulfillment Analytics (Olist) | SQL (DuckDB), Window Functions

- Built reusable order-, customer-, and item-level SQL data marts from 99,441 orders, aggregating to order grain before joining to prevent double-counted revenue and protect KPI accuracy; ran cohort analysis, customer segmentation, and retention/churn analysis surfacing a 3.0% repeat-purchase rate, a BRL 7.98M one-time-buyer segment, and an 8.1% late-delivery rate tied to weaker reviews, delivered as an executive scorecard and dashboard.

Diabetes 30-Day Readmission Risk Modeling | Python, scikit-learn, XGBoost

Sep.–Dec. 2025

- Engineered a 160-feature clinical dataset from 101,766 encounters across 130 U.S. hospitals with patient-level train-calibration-test splits preventing data leakage in a reproducible pipeline.
- Benchmarked logistic regression, random forest, gradient boosting, and XGBoost with cross-validation and hyperparameter tuning for imbalanced classification (AUROC 0.668, AUPRC 0.230); cut Expected Calibration Error 0.342 to 0.004 by isotonic recalibration, added conformal prediction at 90.1% coverage, ran SHAP explainability and subgroup fairness/bias auditing.

LLM Metadata Extraction Pipeline (Kaggle) | Python, NLP, LLM APIs

Jan.–Mar. 2026

- Built a natural language processing (NLP) pipeline extracting structured SDRF metadata across a 71-category taxonomy from 107 publications and 1,659 records; root-caused a 0.575 validation-test gap and benchmarked regex versus LLM structured output via the PRIDE REST API.

TECHNICAL SKILLS

Programming: Python (pandas, NumPy, scikit-learn, XGBoost, SHAP, Matplotlib), R, SQL (PostgreSQL, Oracle SQL, DuckDB), C++, MATLAB, Git version control, Jupyter

Analytics & BI: Tableau, Power BI, Excel (pivot tables), dashboard development, data visualization, KPI reporting, cohort analysis, segmentation, retention/churn analysis

Statistics & ML: Machine learning, statistical modeling, hypothesis testing, statistical significance, A/B testing/experimental design, causal inference, regression, predictive modeling, feature engineering, model evaluation (precision/recall), model calibration, explainability, fairness/bias auditing, time series forecasting (SARIMA, Holt-Winters), optimization

Data Engineering & AI: ETL/ELT pipelines (Mage), Apache Spark, Snowflake, data warehousing, data modeling, normalization, data quality, data cleaning, Natural Language Processing (NLP), Large Language Models (LLMs), LLM APIs, prompt engineering, REST APIs

Business & Communication: Stakeholder communication, cross-functional collaboration, requirements gathering, business acumen, presenting to technical/non-technical audiences, technical writing/documentation, data-driven decisions, working through ambiguity. Languages: Chinese (Native), English (Professional), Japanese (Basic)